

Jiyanshu Singh

B.Tech Biotechnology student at NIT Rourkela | Building ML systems

Contact

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Education

B.Tech Biotechnology · National Institute of Technology, Rourkela

Expected 2027 · CGPA: 7.09/10 · Relevant coursework: Data Structures & Algorithms, Statistics, Mathematics for Engineers

Experience

AI Intern · Digitaaztrans Technologies (Remote) *Jan 2025 – Present*

MLEngineering: Designed, deployed, and operated an ML-filtered multi-strategy trading engine on Google Cloud Platform serving live market data.

Deployment: Containerized the system with Docker; orchestrated 6 parallel strategies on GCP with PostgreSQL logging of 38,500+ labeled trades.

Validation: Applied walk-forward testing across 3 disjoint bull/bear market windows to ensure out-of-sample robustness.

Performance: The ML trade filter improved simulated portfolio from a -■225,000 drawdown to +■93,000 profit.

Freelance Data Scientist / ML Engineer *Dec 2023 – Present*

End-to-end delivery: Built and deployed 4 production web applications (A/B Testing Dashboard, Sales CRM, Invoice Generator, Hydrogel Portal) — all with Dockerized FastAPI/Flask backends on Render/GitHub.

Statistical consulting: Delivered A/B test analysis (p-values, Bayesian Monte Carlo, power analysis) for small clients.

Projects

Sales CRM Dashboard [Code]

Purpose: Full-stack CRM for managing sales leads with KPI dashboard, funnel analysis, revenue charts, CSV export, and basic auth.

Tech stack: FastAPI, Bootstrap 5, Plotly.js, SQLAlchemy, Plotly Dash, Docker, Render.

Invoice Generator [Code]

Purpose: SaaS-ready invoice generation service with PDF export, SQLAlchemy backend, template-based rendering, and Docker deployment.

Tech stack: FastAPI, SQLAlchemy, ReportLab (PDF), Docker, Render.

Skills

Languages: Python, JavaScript, SQL

Backend: FastAPI, Flask, SQLAlchemy, Pydantic, ReportLab (PDF generation)

Frontend: Bootstrap 5, Plotly.js, JINJA2 templates

DevOps: Docker, Docker Compose, Render, GCP, PostgreSQL, Git, REST APIs

Achievements

- Deployed 5 production web applications (FastAPI/Flask + Docker + Render/GCP).
- Labeled 38,500+ trades for ML filter; 95% false-signal reduction; walk-forward OOS validated across 3 market windows.
- 350-sample peptide benchmark with leakage prevention; amyloid ROC-AUC 0.888.

Skills I'm learning: XGBoost, walk-forward testing, Bayesian A/B testing, biotech ML.